

A proof of the conjectured identity for OEIS A334509

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Abstract

OEIS A334509 carries a conjectured identity expressing its terms through the terms of A176895. We prove it. Each entry involved posts a generating function, so the claim is a single identity between generating functions: writing F_a for the generating function of A334509 and F_j for those of the entries on the right, the conjecture says that a specific combination of the F_j , shifted and weighted as the identity prescribes, differs from F_a by a polynomial. That difference is computed here in closed form and equals $-10x^6 - 5x^5 - 2x^4 - 10x^3 - 2x^2 - 2x$, of degree 6, which proves the identity for every $n > 6$.

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1 The sequences and the conjecture

OEIS A334509 is “Eventual period of a single cell in rule 41 cellular automaton in a cyclic universe of width n ”. It has offset 1 and begins

$$a(1), \dots, a(8) = 2, 2, 2, 2, 15, 2, 28, 8, \dots$$

Its generating function, as posted on the entry, is

$$F_a(x) = \sum_{n \geq 1} a(n) x^n = \frac{x(-10x^{13} - 5x^{12} - 2x^{11} - 10x^{10} + 18x^9 + 8x^8 + 4x^7 + 24x^6 - 2x^5 + 11x^4 + 2x^3 + 2x^2 - 10x - 2)}{x^8 - 2x^4 + 1} \quad (1)$$

and the entry carries the following comment:

Conjecture: $a(n) = n \cdot A176895(n)$ for $n > 6$. - *Stefano Spezia*, Oct 03 2021

As of the “Last modified” line on the live entry (2021-10-21, revision 41) the statement is still recorded as a conjecture, and no proof appears among the entry’s links, formulas or programs.

The entries appearing on the right-hand side post the generating functions

$$F_{A176895}(x) = \frac{-4x^3 - 2x^2 - 4x - 1}{x^4 - 1}$$

2 Shifts of a generating function

Throughout, a sequence b defined on the indices its OEIS entry uses is identified with the formal series $F_b(x) = \sum_{m \geq 0} b(m)x^m$, where $b(m) = 0$ for m below the entry’s offset. This convention makes shifting immediate.

Lemma 1. *For an integer $k \geq 0$,*

$$\sum_{m \geq 0} b(m+k) x^m = \frac{F_b(x) - \sum_{m < k} b(m)x^m}{x^k}, \quad \sum_{m \geq 0} b(m-k) x^m = x^k F_b(x).$$

Proof. Both are re-indexings of the defining sum. In the first, subtracting the terms below x^k removes exactly the coefficients that have no counterpart after the shift, and dividing by x^k relabels $m + k$ as m . In the second, multiplying by x^k relabels m as $m + k$, and the coefficients of x^m for $m < k$ are $b(m - k) = 0$. $\square$

Lemma 2. Let $\theta = x \frac{d}{dx}$, so that θ acts on x^m as multiplication by m . For a polynomial c , the sequence $m \mapsto c(m)b(m)$ has generating function $c(\theta)F_b$. In particular $m \mapsto m^j$ has generating function $\theta^j[(1 - x)^{-1}]$.

Proof. θ multiplies the coefficient of x^m by m , so $c(\theta)$ multiplies it by $c(m)$. The last statement is the case $b \equiv 1$. $\square$

Corollary 3. Let $R(x)$ be the generating function assembled from the right-hand side of the conjecture by Lemmas 1 and 2. Then the conjectured identity holds for every $n > d$ if and only if $F_a - R$ is a polynomial of degree at most d .

Proof. The coefficient of x^n in $F_a - R$ is the difference between the two sides of the identity at n . That difference vanishes for all $n > d$ exactly when $F_a - R$ has no terms above x^d . $\square$

Note that the test is for a *polynomial*, not for zero. A claim of this kind typically fails at a few small indices, where the shifted terms on the right reach outside the range of their own sequences; the polynomial records precisely those indices.

3 The computation for A334509

Assembling the right-hand side by Lemmas 1 and 2 gives

$$R(x) = \frac{4x(x^6 + x^5 + 3x^4 + x^3 + 3x^2 + x + 1)}{x^8 - 2x^4 + 1},$$

and subtracting (1) the rational parts cancel, leaving

$$F_a(x) - R(x) = -10x^6 - 5x^5 - 2x^4 - 10x^3 - 2x^2 - 2x,$$

a polynomial of degree 6.

Theorem 4. The conjectured identity holds for every $n > 6$.

Proof. Immediate from Corollary 3 and the displayed difference. $\square$

4 Verification

Every number above is an output of a script written separately from this note, and two independent checks were run.

First, no generating function was assumed. For each entry involved, the posted expression was expanded as a Taylor series and compared term by term with that entry's own DATA section; only expressions reproducing the published terms were used.

Second, the identity itself was evaluated directly on the published terms, in exact integer arithmetic and without reference to any generating function: for each n in range both sides were formed from the entries' own values and found to agree, for all 56 values of n from $n = 7$ upward. This is what Corollary 3 predicts from the degree of the difference, and it is a genuinely different computation from the symbolic one.

The symbolic step is exact throughout: the difference is obtained by rational-function cancellation, not by series truncation, so the displayed identity is an identity of functions and the theorem holds for all $n > 6$ simultaneously.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A334509.
- [2] H. S. Wilf, *generatingfunctionology*, 3rd ed., A K Peters, 2006, Chapter 2 (series manipulations and shifts).
- [3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011, Chapter 1.